

Image Partitioning for Compact Encoding

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Abstract—The exponential growth of medical imaging data necessitates retaining its original diagnostic quality, and for that, some efficient compression methods are applied. The project proposes an ROI-based image compression technique using the wavelet transform for brain tumour images. The image segmentation is Pure ROI, pure non-ROI, and mixed regions. Non-ROI blocks are compressed to minimize the storage and enhance the transmission property, while the ROI blocks are enhanced to preserve the important diagnostic details. The mixed blocks are processed so the inverse wavelet transform handles the ROI part, whereas the wavelet transform processes the non-ROI. The proposed method strives to achieve an optimal balance between the compression efficiency and the diagnostic quality of the image sample. This particular approach is widely used in medical imaging applications where compression is essential for optimizing the storage and transmission while maintaining the diagnostic accuracy.

Index Terms—ROI-based compression, Wavelet transform, Medical imaging, Brain tumor, Image segmentation, Diagnostic quality, Data compression, Image transmission, Storage optimization

I. INTRODUCTION

The production of medical image data has been growing exponentially with the improved technologies of medical imaging such as X-ray, MRI, and CT. These images are often rich in essential and critical diagnostic information that must be stored and transferred at the same quality as the original version of the image. However, coping with bulk data from medical images while retaining diagnostic accuracy can be a significant problem for medical systems. Image compression has therefore become an important solution to the ongoing problem of producing large amounts of data. Image compression will continue to reduce the file size of an image while retaining some of the essential visual and diagnostic information from the original version, while also saving valuable space. Optimally, images will be transmitted more efficiently, increasing the access and work flow for medical systems. Hybrid approaches in compression, particularly those based on Region of Interest (ROI) and Region of Non-Interest (RONI) techniques, preserve important images using lossless techniques for ROI while applying greater compression to RONI. These approaches create an optimal balance of image quality, compression, and reliability for diagnostics which is increasingly valuable for medical imaging today.

II. METHODOLOGY

The proposed methodology for **Region-of-Interest (ROI) Aware Lossy Image Compression** is implemented in the following stages:

A. ROI Detection (Tumor Isolation)

- Calculate the **mean intensity** of the image.
- Create a binary **ROI mask** by setting a very high threshold (e.g., $1.5 \times$ mean intensity) to isolate only the brightest tumor pixels. .

B. Uniform Partitioning

- Divide the image into a **uniform grid** of rectangle blocks (e.g., 10×10) to enable localized adaptive encoding.

C. Block Classification and Compact Encoding

- Each block is classified based on its **ROI pixel ratio** (percentage of ROI pixels in the block).
- **Pure ROI Blocks ($\geq 80\%$ ROI):** Apply **Sharpening** (Unsharp Mask) to maximize detail preservation.
- **Pure RONI Blocks ($\leq 1\%$ ROI):** Apply **Aggressive Lossy Compression** (strong Gaussian Blur + coarse quantization) to achieve higher data reduction.
- **Mixed/Boundary Blocks ($1\% < \text{ROI ratio} < 80\%$):** Apply **Wavelet-like Enhancement** (mild sharpening) to preserve partial ROI content while maintaining compression efficiency.

D. Final Compression

- Reassemble the processed blocks into the final image.
- Save the result using **low-quality JPEG compression** (e.g., Quality = 20) to achieve a high compression ratio while maintaining perceptually important ROI regions.

III. RESULTS

- 1) **Original Image:** The first image represents the uncompressed MRI scan containing both healthy tissues and the tumor region. It serves as the baseline for evaluating visual quality and compression performance.
- 2) **Uniform 100 Rectangles (ROI/RONI/Mixed):** The second image shows the **uniform grid partitioning** of the input image into 10×10 rectangular blocks.
 - **Red blocks** correspond to **ROI regions** containing tumor tissue.
 - **Blue blocks** represent **RONI (Region of Non-Interest)** areas containing normal brain tissue.

- **Green-highlighted blocks** indicate **mixed regions** that partially overlap with the tumor boundary.

This stage confirms the block-wise classification used for adaptive encoding.

3) **ROI-Aware Enhanced Encoding:** The third image shows the final reconstructed image after adaptive processing and compression.

- The **ROI (tumor area)** retains clarity and fine detail due to the application of **sharpening filters**.
- The **RONI regions** appear smoother and slightly blurred, indicating strong lossy compression applied to reduce data size.

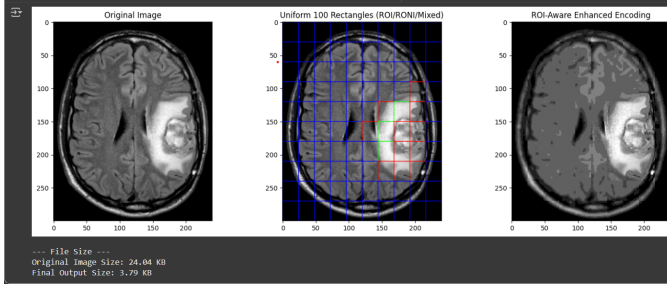


Fig. 1. Result of the code

IV. DISCUSSIONS

Medical images are different from general photos, though, because they include diagnostically important areas where even slight loss of information might cause misunderstandings. As a result, medical image compression techniques need to strike a balance between diagnostic accuracy and data compression. To overcome these difficulties, several strategies have been put forth over time, including transform-based compression, region-based enhancement, and predictive coding.

To reduce redundancy, earlier compression systems such as **JPEG and JPEG-LS**, used entropy and predictive coding. Despite their computational efficiency, these methods **compress all regions equally**, which **may lead to the loss of diagnostic data in ROI**.

Then the **block based PCA method** (Principal Component Analysis) was developed to more effectively take use of local correlations in medical images. It **reduces the redundancy while preserves the required information by converting every image rectangle into a new subspace**. Disadvantage is **lack of spatial-frequency representation** restricts its adaptation to various image topologies, and **uniform block division result in block distortion**.

The region based compression methods were introduced to preserve diagnostic area and accuracy. **JPEG2000 scales and prioritizes ROI coefficient using wavelet transform**. It has drawbacks along with its benefits, when a block has both ROI and non ROI region, boundary distortion is observed. When

the mixed block is compressed all the parameters are applied uniformly **giving blurred or expanded ROI edges**. This could mislead the medical prediction.

The techniques introduced so far are unable to handle the mixed blocks of ROI and non ROI. In our methodology we **divide the image into c rectangles** using rectangular partitioning in order to close this gap. Then further every rectangular block is **divided into 3 categories, Pure ROI, Pure non ROI and the mixed blocks**. Spatial-domain filters are used to **enhance pure ROI regions and blurs the non ROI region to reduce redundancy**, **wavelet and inverse wavelet used for mixed blocks**. The **inverse wavelet is used for ROI reconstruction**, while **wavelet is used for background compression**. This hybrid approach results in good diagnostic fidelity, compact encoding and tumor diagnosing accuracy, free of boundary distortion.

V. CONCLUSION

In conclusion, this project provides a viable and efficient approach for image compression which is based on the ROI (Region of Interest) concept and the wavelet transform. The image is sliced into rectangular blocks based on their taxonomies as pure ROI, pure Non-ROI, and as mixed regions. Each block is processed in terms of significance; including, enhancing ROI blocks to preserve diagnostic or content significance, blurring Non-ROI blocks to gain savings in data storage, and applying the inverse wavelet transform to ROI parts and the wavelet transform to Non-ROI parts for mixed blocks. Overall, this method then compresses the image efficiently by following this selective processing approach to maintain critical image information. An important factor of the project is balancing optimal image quality for important regions while maximising overall image data compression. The project is thus a practical and optimised solution for contemporary image compression while maintaining the necessary key elements required, especially in medical imaging, remote sensing, and securely transmitting images.

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